The empirical recurrence for OEIS A234829: recovering a conjecture that is not written down

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7 September 2026

Abstract

OEIS A234829 records "Empirical recurrence of order 92", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 135$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 92, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 92 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture one cannot read

OEIS A234829 is "Number of (n+1)X(5+1) 0..2 arrays with each 2X2 subblock having the number of clockwise edge increases less than or equal to the number of counterclockwise edge increases.". It has offset 1 and begins

$$212850, 49366557, 10572386472, 2194611652359, 449708506898208, 91647618370398336, 18633171647248503816,$$

The entry states:

> Empirical recurrence of order 92 (see link above)

As of the "Last modified" line on the live entry (Jul 23 2025 08:28:13, revision 7) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 730$ vertices, $a(n) = \iota^\top M^{j(n)}\tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 135$ of the 730, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' =$ 135.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 1460$ exact terms of the model returns order 92 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{92} c_i\, a(n-i),$$

c_1	546	c_{24}	−46721299705340040346398818905	c_{47}	112582541116213
c_2	−121530	c_{25}	25672300016424245577263033 1541	c_{48}	−91094284577071
c_3	15179337	c_{26}	−126259377067337308176707565 3779	c_{49}	230143834255872
c_4	−1219796621	c_{27}	629164108649095991910514569 9014	c_{50}	444217254445345
c_5	68019650875	c_{28}	−3401489065349371882867169751 6244	c_{51}	−71881485027243
c_6	−2757702507876	c_{29}	1826186905120854054284400178 49699	c_{52}	560752780452096
c_7	83738969197083	c_{30}	−8689926698976186044515159772 94738	c_{53}	−23199817886337
c_8	−1938582670772926	c_{31}	3460120450253134483480364761 542966	c_{54}	−4449306366702
c_9	34457299572027394	c_{32}	−1125177127883436943765341454 1250698	c_{55}	80023532379202
c_{10}	−467354235142970254	c_{33}	2918174924332495532715738863 2086719	c_{56}	−7357907781551
c_{11}	4692782392108901385	c_{34}	−5687662681407144079444423066 3224894	c_{57}	64310507086744
c_{12}	−31566080747912967059	c_{35}	6540346787819080592017783097 9934385	c_{58}	−6646898182764
c_{13}	80524158096204057584	c_{36}	4878224200710532454173693865 5523278	c_{59}	62416323783282
c_{14}	106639079295762843 3228	c_{37}	−5175027661592972139897550802 72013409	c_{60}	−4551595573663
c_{15}	−1687458920947435146 4682	c_{38}	1706558957009970553331467819 712041164	c_{61}	24674931936537
c_{16}	1280673711907335802 71301	c_{39}	−3988009802968736844350514573 276343582	c_{62}	−9352014255216
c_{17}	−576723830570402953 110250	c_{40}	7429294731008202171969712881 295113304	c_{63}	17971323682713
c_{18}	100520664562073888 8069816	c_{41}	−1139016371448136762713838248 3275012982	c_{64}	5286088842784
c_{19}	640496099988570510 7032265	c_{42}	1432444039068194280591510447 3974629254	c_{65}	−660222612047
c_{20}	−642246391204226788 73646897	c_{43}	−1419144183398921600887333239 1122502666	c_{66}	3266541872062
c_{21}	324552257088487442 545003863	c_{44}	9643547735481623515817089236 487260201	c_{67}	−987561699425
c_{22}	−142819069652157751 0893202269	c_{45}	−1468392572488837532968831239 530813377	c_{68}	1527189052832
c_{23}	769256343816156460 2295774872	c_{46}	−6957838644957735560872991247 085194364	c_{69}	137674699847

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 92, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $92 + 4$ terms removed returns order 92 as well, so $\deg q_{\min} = 92 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. □

4 Verification

Three checks.

First, the walk model reproduces all 10 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $92 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 92 that this sequence satisfies — and that family turns out to have exactly one member.

References

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